Padding

Padding = is the technique of adding extra pixels (usually zeros) around the border of an input image or feature map before convolution. it is primarily used to prevent spatial dimensions from shrinking, retain boundary information, and allow for deeper network architectures.

It is done to reduce the loss of data among the side/boundary of the image
Half padding = half of feature size
Full padding = it equal to size of the filter/kernel

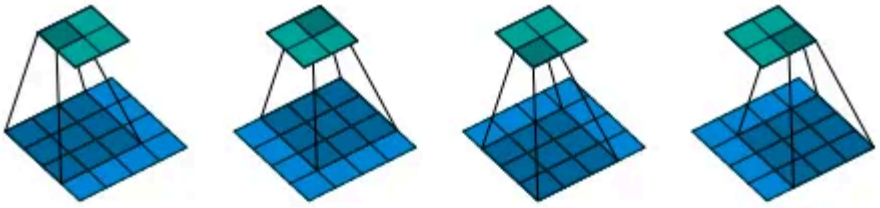

Figure 2.1: (No padding, unit strides) Convolving a 3×3 kernel over a 4×4 input using unit strides (i.e., $i = 4$, $k = 3$, $s = 1$ and $p = 0$).

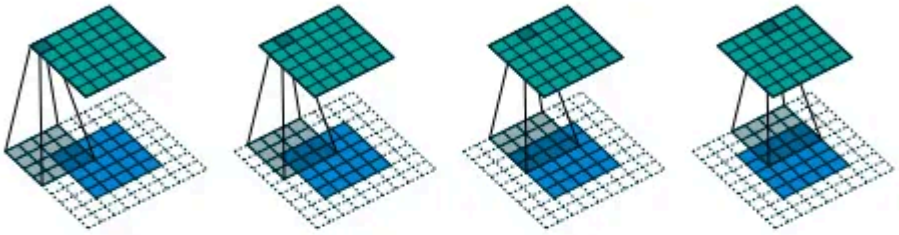

Figure 2.2: (Arbitrary padding, unit strides) Convolving a 4×4 kernel over a 5×5 input padded with a 2×2 border of zeros using unit strides (i.e., $i = 5$, $k = 4$, $s = 1$ and $p = 2$).

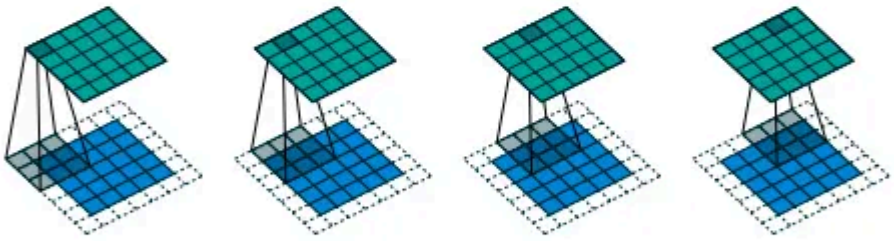

Figure 2.3: (Half padding, unit strides) Convolving a 3×3 kernel over a 5×5 input using half padding and unit strides (i.e., $i = 5$, $k = 3$, $s = 1$ and $p = 1$).

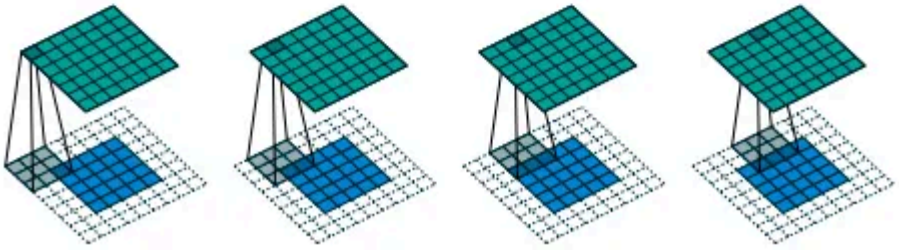

Figure 2.4: (Full padding, unit strides) Convolving a 3×3 kernel over a 5×5 input using full padding and unit strides (i.e., $i = 5$, $k = 3$, $s = 1$ and $p = 2$).

This can be done in 2 ways:

- **Valid Padding:** no padding is added to the input feature map, and the output feature map is smaller than the input feature map. this is useful when we want to reduce the spatial dimensions of the feature map.
- **Same Padding:** in the same padding, padding is used to the input feature map such that the size of the output feature map is the same as the input feature map. this is useful when we want to preserve the spatial dimensions of the feature maps.

The most commonly used approach is **zero-padding** because of its performance, simplicity, and computational efficiency.
this technique involves adding zero symmetrically around the edges of the an input. this approach is adopted by many high-performing CNNs such as AlexNet.

Geek4Geek